

Supplement to AWR2944EVM Radar Software Installation Manual

This document serves as a technical supplement to the AWR2944EVM DSS Program Manual. It provides a recommended execution order, explicit tool dependency requirements, environment discovery sequences, path-sensitivity guidelines, and build fixes associated with the [RPS_Fix_Compilation branch on GitHub \(Fontys-engineering/Radar_software\)](https://github.com/Fontys-engineering/Radar_software).

1. Revised Reading & Execution Order Guide

To ensure a smooth setup without running into environment conflicts or hardware connection issues, execute the setup in the exact order detailed below:

1. Environment & Toolchain Setup: Install Code Composer Studio (CCS), SysConfig, mmWave MCU+ SDK, and UniFlash in the specified order before launching CCS or connecting hardware.
(Sections 2 & 3 of this Supplement)
2. Repository Setup & Path Verification: Clone the fixed branch into the expected root path before opening or configuring workspace settings.
(Section 4 of this Supplement)
3. Hardware & Debug Firmware Setup: Configure board jumpers, connect power/USB, and flash the CCS debug image via UniFlash.
(Original Manual: Steps 1 to 4)
4. Importing, Building & Debugging: Load the workspace in CCS, build projects, connect target cores, and upload MSS/DSS binaries.
(Original Manual: Steps 5 to 6 + Section 5 of Supplement)
5. System Verification: Confirm core execution and verify active data stream in the Demo Visualizer.
(Section 6 of this Supplement)

2. Dependency Matrix & Explicit Tool Versions

Install the software tools using the exact versions listed below. Note that SysConfig 1.28 is strictly required; earlier recommendations to force version 1.23 no longer apply.

Tool Name	Version	Function	Recommended Install Path
Code Composer Studio	20.4.1	IDE & Debugger	C:\ti\ccs2041
SysConfig	1.28	Pin and Peripheral Configurator	C:\ti\sysconfig_1.28.0
mmWave MCU+ SDK	4.7.2.1	Board SDK	C:\ti\mmwave_mcuplus_sdk_04_07_02_01
Uniflash	8.6.0	Firmware flasher	C:\ti\uniflash_8.6.0

3. Environment Installation & Tool Discovery Sequence

Install all prerequisites prior to opening CCS to ensure the IDE discovers installed SDKs and configuration tools automatically:

1. Sequential Installation: Install CCS 20.4.1, SysConfig 1.28, mmWave MCU+ SDK 4.7.2.1, and UniFlash directly into C:\ti\.
2. Tool Discovery in CCS:
 - Launch CCS 20.4.1.
 - Navigate to File -> Preferences -> Code Composer Studio -> Products.
 - Click the Refresh / Discover button next to *Discovered products*.
 - Confirm that SysConfig 1.28 and mmWave MCU+ SDK 04.07.02.01 appear in the discovered products list.
 - Click Apply and Close, then restart CCS.

4. Path Sensitivity & Repository Location

The CCS project and workspace configurations rely on hardcoded paths referencing C:\Git\Radar_software.

- Default Path: Clone or extract the repository directly into C:\Git\Radar_software.
- Alternative Working Directory: If using a custom directory (e.g., D:\Projects\Radar_software), you must manually update the project macros:
 - a. In the CCS Project Explorer, right-click Radar_Program_dss -> Properties.
 - b. Navigate to Build -> Variables.
 - c. Update PROJECT_ROOT and SDK search paths to reflect your local path.
 - d. Repeat the procedure for Radar_Program_mss.

5. Build & Dependency Fixes (RPS_Fix_Compilation Branch)

When building the project, ensure you checkout the [RPS_Fix_Compilation](#) branch of the repository, which addresses compiler and dependency issues present in earlier releases:

- SDK Inclusion Linkage Fix: Previously, modified SDK header and source files were placed in the local [Dependencies](#) directory, but compiler preprocessors defaulted to using unmodified files from the globally installed SDK in `C:\ti\`. The [RPS_Fix_Compilation](#) branch bundles a locally modified copy of the SDK into the project tree and updates inclusion flags (`-I`) so local files compile and link correctly.
- Single SysConfig Tooling: Removes dual-version SysConfig requirements, standardizing the build pipeline strictly on SysConfig 1.28.
- Local Dependency Bundling: Dependency files are included locally within the branch repository. Host installations of CCS 20.4.1, mmWave MCU+ SDK 4.7.2.1, and SysConfig 1.28 remain required to supply base toolchains ([ti-arm-clang](#), [C6000 CGT](#)).

6. Final State Verification ("What Success Looks Like")

Your environment and board setup are complete and functioning correctly when the following criteria are met:

1. Build Verification

In the CCS Project Explorer after selecting Project $\rightarrow$ Build All (Ctrl+B):

- **Radar_Program_mss** -> Generated Binaries contains **Radar_Program_mss.xer5f**.
- **Radar_Program_dss** -> Generated Binaries contains **Radar_Program_dss.xe66**.
- The CCS Console output concludes with **0 Errors**.

2. Execution & Core State Verification

With SOP jumpers set to Functional Mode (**001**) and project-less debug started (AWR2944.ccxml):

- **Cortex_R5_0** core shows state Running after loading **Radar_Program_mss.xer5f** and resuming.
- **C66xx_DSP** core shows state Running after loading **Radar_Program_dss.xe66** and resuming.

3. Demo Visualizer Connection

When connecting through the TI mmWave Demo Visualizer:

- **CFG_port** (User UART): Configured to XDS110 Class Application/User UART.
- **DATA_port** (Auxiliary): Configured to XDS110 Class Auxiliary Data Port.
- **Baud Rate**: Set to Custom (3125000).
- **Expected Result**: Bottom status bar displays "Hardware Connected" in green. Clicking Send Config to mmWave Device renders real-time point-cloud and Doppler plots under the Plots tab.